Third Eye — Requirements Specification

Project: Third Eye
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Source of truth: `Third_Eye_Project_Plan.md` (plan of 2026-09-18). Every requirement below traces to a plan section; where this document and the plan disagree, the plan wins and this document is a bug.

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1. Purpose and scope

1.1 Purpose

This document turns the Third Eye project plan into a numbered, testable set of requirements. It exists so that (a) each module owner knows exactly what "done" means for their module, (b) the evaluation in week 11 has pre-agreed pass/fail criteria rather than post-hoc ones, and (c) procurement and environment setup in week 1 is a checklist rather than a scramble.

1.2 Product scope

Third Eye observes a person performing a physical procedure through one or more fixed cameras and raises real-time alerts when a step is missed, performed out of order, or performed incorrectly. The procedure itself is not hard-coded: an LLM compiles a written manual or rulebook into a machine-readable **procedure spec**, and a deterministic checker compares observed world state against that spec.

Two procedures are in scope:

Procedure	Spec form	Role
LEGO assembly (Model A, Model B)	Dependency graph (DAG)	Must-have demo and primary training domain
Jenga (game rules)	Constraint set, no fixed order	Generality result and live adversarial demo

1.3 Document scope

Covered: functional, non-functional, data, hardware, software, interface, evaluation and milestone-gate requirements, plus the open items that block requirements from being finalised.

Not covered: implementation design (that is plan §2 and §4), work allocation (plan §9), and the report/poster content (plan §11).

1.4 Definitions

Term	Meaning
Procedure spec	JSON document compiled from a manual/rulebook; either DAG form (§4.2 of plan) or constraint form (§4.3)
World state	Current physical configuration as tracked by the system: LEGO stud grid, or Jenga 18×3 lattice
Event	A discrete observed change (`ADD` , `REMOVE` , `STEP_START` , `STEP_END`) emitted by the state tracker or the temporal model
Checker	Deterministic engine that evaluates spec preconditions against events and raises alerts

Term	Meaning
Hard dependency	Physically necessary precondition (support). A violation indicates *our* perception failed, never a user error
Soft dependency	Order stated by the manual. A violation is a genuine out-of-order user error
Occlusion window	Interval during which a hand covers the workspace and the state tracker is frozen
Tier 1 / 2 / 3	The generality ladder of plan §1.2: unseen assembly / unseen domain / unseen procedure topology

1.5 Identifier and priority scheme

Requirement IDs are `<CLASS>-<MODULE>-<n>` and are stable — never renumber, mark superseded instead.

Class	Meaning
`FR`	Functional requirement
`NFR`	Non-functional requirement
`DR`	Data requirement
`HW`	Hardware requirement
`SW`	Software / environment requirement
`IF`	Interface / contract requirement
`EV`	Evaluation acceptance criterion
`MG`	Milestone gate

Priority uses MoSCoW, aligned to plan §6:

Priority	Meaning
M	Must have — the minimum viable project. Failure here fails the project
S1–S6	Should have, in the plan's priority order. Delivered only once all **M** items work end to end
C	Could have / stretch. Attempted only if week 13 slack is genuinely clear
W	Won't have this semester — recorded so it is not silently re-scoped in

Verification method per requirement: **T** = automated test, **D** = live demonstration, **A** = analysis / measurement against recorded data, **I** = inspection or review.

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2. Stakeholders

Stakeholder	Interest	Requirements they sign off
Course panel / evaluators	Working live demo, defensible results	FR-UI, EV-*, MG-*
Perception lead	Calibration, detector, tracking, both state trackers	FR-CAL, FR-PER, FR-STA
Temporal model lead	Features, MS-TCN++, benchmark scripts	FR-TMP, EV-SEG
Reasoning lead	Schemas, compiler, validator, checker, explainer	FR-CMP, FR-CHK, FR-EXP, IF-*
App lead	Streaming backend, UI, integration, latency	FR-UI, NFR-PERF, IF-WS
Demo participants (panel member, teammates)	Can operate it without training	NFR-USE
Recorded subjects	Consent and release terms for published data	NFR-PRIV, DR-REL

3. Functional requirements

3.1 FR-CMP — LLM procedure compiler (plan §2.1, §4.2, §4.3)

ID	Requirement	Pri	Verify
FR-CMP-1	The compiler shall accept a plain-text manual (one instruction per step) and emit a procedure spec in **DAG form** conforming to the §4.2 schema.	M	T
FR-CMP-2	For each step the compiler shall emit: step id, verbatim instruction, part descriptor (colour + size), target pose (x, y, layer, rotation) on the stud grid, `hard_depends_on`, `soft_depends_on`, `optional`, and an expected duration range.	M	T
FR-CMP-3	The compiler shall distinguish **hard** (physical support) from **soft** (manual-stated order) dependencies and record both independently.	M	T
FR-CMP-4	The compiler shall accept a rulebook and emit a procedure spec in **constraint form**: one or more action schemas, each with preconditions (`expr` , human-readable `text` , `violation` code) and effects, plus terminal conditions.	S5	T
FR-CMP-5	Precondition expressions shall be drawn from a **fixed predicate vocabulary supplied in the prompt** and evaluated by a small safe interpreter. Python `eval` or any equivalent arbitrary-code path shall not be used.	S5	I, T
FR-CMP-6	The validator shall reject a spec that fails any of: JSON-schema conformance, dependency-cycle check, collision check (two parts occupying the same cells), support check (no floating parts), and — for constraint specs — a satisfiability check that at least one legal action exists from the initial state.	M	T
FR-CMP-7	On a validation failure the system shall report the failing check and the offending step/ constraint id, and shall permit **one automated repair round** through the LLM before requiring human intervention.	M	T
FR-CMP-8	The system shall provide a **teach-by-demonstration** path: record one correct performance, capture the world state after each action, and use the LLM only to name the resulting steps.	S6	D
FR-CMP-9	The UI shall allow a human to edit and re-validate a compiled spec before a session runs.	S6	D
FR-CMP-10	Compilation shall be an offline, once-per-procedure operation. No LLM call shall occur on the per-frame or per-event path.	M	I
FR-CMP-11	The LLM shall never be given raw video or video-derived imagery. Its inputs are text only.	M	I

3.2 FR-CAL — Calibration and cameras (plan §2.2)

ID	Requirement	Pri	Verify
FR-CAL-1	The system shall detect four printed ArUco markers at the baseplate corners and compute a homography from image pixels to **stud coordinates**.	M	T, D
FR-CAL-2	Calibration shall succeed from any reasonable top-down camera angle and shall not require the camera to be perfectly perpendicular.	M	D
FR-CAL-3	The homography shall be recomputed every 10 frames, or immediately when marker positions shift beyond a threshold.	M	T
FR-CAL-4	A manual four-corner click fallback shall be available when marker detection fails.	S1	D
FR-CAL-5	The system shall support a **side camera** calibrated against the same world frame, used as the primary height signal for LEGO.	S1	D
FR-CAL-6	The system shall support **two side cameras at 90°** for Jenga, calibrated against table markers plus known block dimensions (75 × 25 × 15 mm) for scale.	S4	D
FR-CAL-7	The system shall detect and report loss of calibration (markers occluded or out of frame) rather than silently emitting wrong coordinates.	S1	T

3.3 FR-PER — Perception (plan §2.3)

ID	Requirement	Pri	Verify
FR-PER-1	The system shall detect hand keypoints every frame and derive: hand-over-workspace, grasp, and **number of hands in contact**.	M	T
FR-PER-2	The system shall detect parts every frame with a fine-tuned YOLO detector, classifying LEGO parts by colour and size (`red_2x4` , `blue_2x2` , `yellow_1x4` , …).	M	T
FR-PER-3	A zero-shot open-vocabulary detector shall be available as a documented fallback and as the ablation baseline.	M	A
FR-PER-4	The system shall maintain stable part identities across frames using a multi-object tracker, so that part motion can be attributed.	M	T
FR-PER-5	The Jenga detector shall use a **single part class**, so that a Jenga failure is attributable to state or reasoning logic rather than to detection.	S4	I
FR-PER-6	While a hand occludes the workspace the world state shall be **frozen**; updates resume only after the hand has left.	M	T

3.4 FR-STA — World-state tracker (plan §2.4)

ID	Requirement	Pri	Verify
FR-STA-1	The world-state tracker shall be an **interface** exposing `current()` and `diff()` , with LEGO and Jenga as independent implementations. No checker code shall branch on procedure type.	M	I, T
FR-STA-2	The LEGO tracker shall maintain the model as a list of placed bricks `(type, x, y, layer, rotation)` in stud coordinates.	M	T
FR-STA-3	A brick shall be counted as placed only after it has remained still on the plate for ≈0.5 s following a hand exit.	M	T
FR-STA-4	The tracker shall emit a state diff after each placement or removal as an event, e.g. `ADD red_2x4 @ (0,0,L0,rot0)` .	M	T
FR-STA-5	Layer (height) shall be inferred from top-down occupancy **only** where no side camera is available; once integrated, the side camera shall be the primary height signal.	S1	A
FR-STA-6	The tracker shall support a **forced re-verification** request for named prior steps, resolved once the workspace view is clear (see FR-CHK-7).	M	T
FR-STA-7	The Jenga tracker shall maintain an 18 × 3 occupancy array plus per-layer orientation, derived from two orthogonal side views.	S4	T
FR-STA-8	The Jenga tracker shall emit `ADD (layer, slot)` and `REMOVE (layer, slot)` events, and `COLLAPSE` as a terminal event detected from a large single-frame drop in tower height.	S4	T, D

3.5 FR-TMP — Step recognizer (plan §2.5)

The step recognizer is a **benchmark-only track, time-boxed to weeks 5 and 9**. It is not on the live critical path for either demo. Any requirement here that threatens a Must-have requirement is cut, not rescheduled.

ID	Requirement	Pri	Verify
FR-TMP-1	The system shall extract clip-level features from a pretrained video encoder plus perception features (hand positions, part-in-hand, part counts), and cache them per video.	M	T
FR-TMP-2	A temporal action segmentation model shall label each moment with a generic action class: `search` , `grasp` , `place` , `adjust` , `remove` , `idle` .	M	T
FR-TMP-3	The model shall predict **generic verbs, not step ids**, so that it transfers to Model B and to Jenga without retraining. Step identity comes from the world-state tracker.	M	I
FR-TMP-4	The temporal model shall run asynchronously at ≈2 Hz and shall never block a video frame.	M	T, A

ID	Requirement	Pri	Verify
FR-TMP-5	The temporal model shall supply the **only** action signal during occlusion windows, and its events shall be consumable by the checker on the same interface as state events.	S	T
FR-TMP-6	The system shall run the step recognizer standalone on a public benchmark that has no stud grid.	M	A

3.6 FR-CHK — Checker (plan §2.6, §4.4)

ID	Requirement	Pri	Verify
FR-CHK-1	The checker shall maintain per-step status in { `pending`, `active`, `done`, `error` } and evaluate preconditions on every received event.	M	T
FR-CHK-2	The checker's native interface shall be the **constraint form**; DAG steps shall be lowered into it so both procedures execute identical engine code.	M	I, T
FR-CHK-3	The checker shall raise **missed step**: a required, non-optional step still `pending` at session end, or implied by a soft-dependency violation.	M	T, D
FR-CHK-4	The checker shall raise **out of order**: a placement matching step *k* while a step it softly depends on is still `pending`.	M	T, D
FR-CHK-5	The checker shall raise **wrong brick**: a placed part whose type matches no pending step at that pose.	M	T, D
FR-CHK-6	The checker shall raise **wrong position / rotation** (type matches a pending step but pose is off by ≥1 stud or 90°) and **extra / removed brick**.	S2	T, D
FR-CHK-7	An unmet **hard** dependency shall **never** produce a user-facing alert. It shall defer the decision, request state re-verification, and resolve as either a logged **perception miss** (part was present, we missed it) or a genuine **missed step** alert.	M	T
FR-CHK-8	Perception misses recorded by the re-verification path shall be logged separately and counted against detector recall in the evaluation, not against the user.	M	A, I
FR-CHK-9	Events below a configurable confidence threshold shall produce a **soft (yellow) warning** rather than a red alert.	S2	T, D
FR-CHK-10	For Jenga the checker shall raise: **illegal source layer**, **layer started early**, **two-handed move**, **wrong replacement**, and **collapse** (terminal).	S5	T, D
FR-CHK-11	Every alert shall be attributable to one explicit rule or precondition id. No alert shall originate from a learned model's output alone.	M	I
FR-CHK-12	The checker shall write a complete, timestamped event and alert log for every session.	M	T

3.7 FR-UI — Interface, streaming and explainer (plan §2.7, §4.5)

ID	Requirement	Pri	Verify
FR-UI-1	The live view shall show the video with part bounding boxes and a grid (LEGO) or lattice (Jenga) overlay.	M	D
FR-UI-2	The live view shall show a **checklist** that ticks steps off as they complete (LEGO) or a **live legality panel** (Jenga).	M	D
FR-UI-3	Alerts shall appear in the live view distinguishing red (error) from yellow (soft warning), with the human-readable rule text.	M	D
FR-UI-4	The demo process shall read the camera **locally in the Python process** and push only overlays and alerts over the WebSocket. Browser-side frame capture may remain available for remote use but is not the demo path.	M	I, A
FR-UI-5	A **ghost overlay** shall indicate the next expected placement.	S6	D
FR-UI-6	Session start, stop and procedure selection shall be operable from the UI without editing code or config files.	S	D

ID	Requirement	Pri	Verify
FR-EXP-1	At session end (or on `COLLAPSE`) the system shall pass the event log to the LLM and produce a readable session report citing timestamps, the step involved and the consequence.	S6	D
FR-EXP-2	The session report shall be generated from the event log only — never from video.	S6	I

3.8 FR-SYS — System-level behaviour

ID	Requirement	Pri	Verify
FR-SYS-1	The end-to-end path camera → perception → state → checker → alerts shall run as a single live session with no manual step between stages.	M	D
FR-SYS-2	Procedure, camera and model choices shall be set by YAML config; switching between Model A, Model B and Jenga shall require no code change.	S	I, D
FR-SYS-3	A session shall recover from transient detector failure (dropped frames, momentary marker loss) without terminating.	S	T
FR-SYS-4	The system shall run entirely offline except for the offline compile and post-session explain calls; a local LLM fallback shall be documented.	S	I

4. Non-functional requirements

4.1 NFR-PERF — Performance and latency (plan §3.1)

Budget is measured on the demo laptop in week 6 and re-measured in week 11.

ID	Requirement	Target	Pri	Verify
NFR-PERF-1	End-to-end live frame rate	≥ 15 FPS sustained	M	A
NFR-PERF-2	Per-frame critical path total	≤ 40 ms (66 ms available)	M	A
NFR-PERF-3	Part detector, per frame	15–25 ms	M	A
NFR-PERF-4	Hand keypoints, per frame	5–10 ms	M	A
NFR-PERF-5	Tracker, per frame	< 2 ms	M	A
NFR-PERF-6	Homography refresh, amortised	< 1 ms	M	A
NFR-PERF-7	State update + diff, per hand-exit	< 1 ms	M	A
NFR-PERF-8	Checker, per event	< 1 ms	M	A
NFR-PERF-9	Temporal model, async, off critical path	≤ 200 ms per 0.5 s window	M	A
NFR-PERF-10	WebSocket overlay push	≤ 10 ms per frame	M	A
NFR-PERF-11	Median delay from error occurrence to alert shown	< 2 s	M	A

4.2 NFR-* — Other quality attributes

ID	Attribute	Requirement	Pri	Verify
NFR-REL-1	Reliability	A full Model A session (12–15 steps) shall complete without a crash or a manual restart in ≥ 9 of 10 consecutive runs.	M	A
NFR-REL-2	Quietness	≤ 1.0 false alerts per fully correct Model A build.	M	A
NFR-USE-1	Usability	A person who has not used the system before shall complete a guided Model A build using only on-screen information.	S	D
NFR-USE-2	Usability	Alert text shall name the step and the physical consequence in plain language, not a rule id alone.	S	I
NFR-MNT-1	Maintainability	Adding a new procedure of an existing state model shall require only a new manual and a compiled spec — no code change.	M	D
NFR-MNT-2	Maintainability	Adding a new **state model** shall require implementing the `WorldState` interface only; checker and compiler code shall be untouched.	S	I
NFR-PORT-1	Portability	Training shall be runnable on a free-tier cloud GPU (Colab/Kaggle) via precomputed features and small model variants.	M	D
NFR-PORT-2	Portability	Inference shall run on a single consumer GPU; a documented CPU-only degraded mode is a stretch goal.	M / C	A
NFR-COST-1	Cost	Total LLM API spend shall stay negligible (≈20 calls across the project); a local model fallback shall be documented.	M	A
NFR-PRIV-1	Privacy	Recorded subjects shall give written consent covering research use and, separately, public release. Faces need not be captured — frame the workspace, not the person.	M	I
NFR-PRIV-2	Privacy	Secrets (LLM API keys) shall live in environment variables or an untracked `.env`; no key shall be committed.	M	I, T
NFR-REP-1	Reproducibility	Every reported number shall be regenerable from a committed script, a fixed data split and a recorded config.	M	I

ID	Attribute	Requirement	Pri	Verify
NFR-REP-2	Reproducibility	Dataset splits shall be fixed in week 8 and never adjusted after a result is reported from them.	M	I
NFR-TRC-1	Traceability	Every experiment run shall be logged to the experiment tracker with its config hash and git commit.	S	I

5. Data requirements (plan §5)

5.1 Public datasets

ID	Requirement	Pri	Verify
DR-PUB-1	Access, licence and storage size for every candidate public dataset shall be **verified in week 1** — the plan's descriptions are from memory and are unverified.	M	I
DR-PUB-2	One public procedural-error dataset with mistake labels shall be secured as the primary benchmark (first choice: a multi-view assembly dataset with fine-grained action and mistake labels).	M	I
DR-PUB-3	At least one alternative benchmark shall be identified and kept viable in case access is denied or slow.	M	I
DR-PUB-4	Any third benchmark is a stretch item and shall not be scheduled.	C	—

5.2 THIRDEYE-LEGO (own recordings)

ID	Requirement	Pri	Verify
DR-LEG-1	**Model A**: 12–15 steps, designed **nearly flat (2 layers)** so that the must-have demo never depends on height inference.	M	I
DR-LEG-2	**Model B**: 10–12 steps, 3–4 layers, built from the same ~8–10 brick types, used **only for testing** (zero-shot, tier 1).	S3	I
DR-LEG-3	Brick vocabulary: ~8–10 standard types (2×2, 2×4, 1×4, 1×2) in clearly distinct colours. No tiny pieces (1×1 plates, Technic pins), no hinged or angled parts.	M	I
DR-LEG-4	Model A: **40–60 runs** from 4–6 people, roughly half containing planted errors.	M	I
DR-LEG-5	Model B: **15–20 runs**, test split only, never used for training or tuning.	S3	I
DR-LEG-6	Recording shall **start in week 4** and accumulate continuously, not begin in week 7.	M	I
DR-LEG-7	Planted error types shall cover all eight: skip a step; swap two steps; wrong colour; wrong size; off-by-one-stud position; 90° rotation; extra brick; remove a placed brick.	M	I
DR-LEG-8	Labels: step start/end times and error type per run; final state per step (from teach-by-demonstration); bounding boxes on 300–600 frames.	M	I
DR-LEG-9	Labelling budget: **30 person-hours**, shared across the team in week 8.	M	A

5.3 THIRDEYE-JENGA (own recordings)

ID	Requirement	Pri	Verify
DR-JEN-1	15–20 games from 3–4 people, recorded from two orthogonal side views.	S4	I
DR-JEN-2	Planted violations shall cover: pull from the top complete layer; start a new layer early; two-handed pull; return a block to an illegal slot. Collapses occur naturally and need no planting.	S5	I
DR-JEN-3	Labels: move start/end, source and destination slot, violation type. Budget **8 person-hours**.	S4	A
DR-JEN-4	Detector training set: ~150 labelled frames, single class, against a dark backdrop.	S4	I

5.4 Compiler eval corpus

ID	Requirement	Pri	Verify
DR-CRP-1	**15–20 short written manuals** for models that are never physically built, each with a hand-authored ground-truth spec. Authored in **week 2**.	M	I

ID	Requirement	Pri	Verify
DR-CRP-2	At least 3–4 corpus entries shall be rulebook-style sources, so that the constraint form is measured and not only the DAG form.	S5	I
DR-CRP-3	Compiler errors shall be reported under an explicit taxonomy: wrong pose, missed dependency, hallucinated step, wrong constraint.	M	A
DR-CRP-4	This corpus is the one result that survives independently of the vision stack, and shall be completed even if perception slips.	M	I

5.5 Release

ID	Requirement	Pri	Verify
DR-REL-1	The compiler eval corpus is text-only and shall be released unconditionally with the code.	M	I
DR-REL-2	Video datasets shall be released only with the consent of everyone recorded (NFR-PRIV-1).	S	I

6. Hardware requirements (plan §3.1, §2.2)

Procurement checklist for **week 1**. Items marked M block the must-have demo.

ID	Item	Spec	Qty	Pri
HW-1	Top-down camera	1080p webcam or phone with a stable USB/IP feed	1	M
HW-2	Overhead camera stand / rig	Rigid, repeatable position; must not drift during a session	1	M
HW-3	LEGO baseplate	16×16 or 32×32 studs	1–2	M
HW-4	Printed ArUco markers	4, mounted at baseplate corners, rigid and non-glossy	1 set	M
HW-5	LEGO brick set	~8–10 distinct types (2×2, 2×4, 1×4, 1×2) in clearly separable colours; enough duplicates for Model A and Model B	1	M
HW-6	Plain mat and even lighting	Matte, non-reflective; consistent illumination across the plate	1	M
HW-7	Laptop with GPU	Runs the full live pipeline within the §4.1 budget; remote GPU acceptable for training only	1	M
HW-8	Side camera	Same class as HW-1; primary LEGO height signal	1	S1
HW-9	Second side camera	Mounted at 90° to HW-8 for Jenga full observability	1	S4
HW-10	Jenga set	Standard 54 blocks, 75 × 25 × 15 mm	1	S4
HW-11	Dark backdrop	High contrast behind the Jenga tower	1	S4
HW-12	Table markers for Jenga	Around the tower base, for side-view calibration and scale	1 set	S4

ID	Constraint	Pri
HW-C1	A minimum of **2 cameras** is required for Jenga; **3** is ideal for the full system. The achievable mount and sync count is an open item (§9).	S4
HW-C2	All cameras shall be time-aligned well enough that a single move is attributable to one timestamp within one frame interval.	S4
HW-C3	Camera positions shall be reproducible between sessions, or recalibration shall be part of the documented session startup.	M

7. Software and environment requirements (plan §3)

7.1 Module → technology commitments

ID	Module	Primary	Fallback	Trained?
SW-1	Calibration	OpenCV ArUco + homography	Manual 4-corner click	No
SW-2	Hands	MediaPipe Hands	MMPose hand models	No
SW-3	Part detection	YOLO (Ultralytics), fine-tuned	YOLO-World zero-shot	**Yes** — 300–600 LEGO frames, ~150 Jenga
SW-4	Tracking	ByteTrack	SORT	No
SW-5	World state	Own grid / lattice modules (NumPy)	—	No
SW-6	Video features	VideoMAE or I3D, precomputed	DINOv2 per-frame	No
SW-7	Step segmentation	MS-TCN++	ASFormer	**Yes** — main temporal model
SW-8	LLM (compiler, explainer)	Claude Sonnet 5 via API	Local Qwen or Llama via Ollama	No — prompt + JSON schema
SW-9	Checker	Plain Python state machine	—	No
SW-10	Backend	FastAPI + WebSocket streaming	—	—
SW-11	Frontend	React	Streamlit for early versions	—
SW-12	Labelling	CVAT (boxes, step segments)	Label Studio	—
SW-13	Experiment tracking	Weights & Biases	TensorBoard	—

7.2 Environment requirements

ID	Requirement	Pri	Verify
SW-E1	The project shall pin a reproducible Python environment (lockfile committed) in **week 1**; the versions in §7.3 are indicative floors, not the final pins.	M	I
SW-E2	Python 3.11+ with PyTorch and CUDA matched to the available GPU driver.	M	T
SW-E3	`pip install -r requirements.txt` (or the chosen lock tool's equivalent) on a clean machine shall bring up a working inference environment. Verified once per semester on a second machine.	M	T
SW-E4	Training-only dependencies shall live in a separate extras group so the demo machine installs the minimum.	S	I
SW-E5	The repository shall follow the directory layout of plan §4.1.	M	I
SW-E6	Unit tests shall exist for the stud grid, the Jenga lattice, the state diff and every checker rule.	M	T
SW-E7	Data, weights and videos shall not be committed to git; a documented storage location and a fetch script shall be provided instead.	M	I

7.3 Dependency groups (indicative; pin in week 1)

Runtime — live demo

```
numpy
opencv-contrib-python        # ArUco lives in contrib
torch                        # CUDA build matched to the local driver
ultralytics                  # YOLO detector + ByteTrack
mediapipe                    # hand keypoints
pydantic                     # procedure-spec schemas
fastapi
uvicorn[standard]            # WebSocket streaming
websockets
pyyaml                       # configs
anthropic                    # LLM compiler + explainer
python-dotenv                # API key, never committed
```

Training and evaluation (extras)

```
torchvision
transformers                 # VideoMAE features
scikit-learn
pandas
matplotlib
wandb
tqdm
```

Development (extras)

```
pytest
ruff
mypy
```

Frontend: Node 20+, React, a WebSocket client. Streamlit is acceptable for pre-week-12 internal views only.

External services: an LLM API key (≈20 calls total) with Ollama + a local model as the offline fallback; a labelling server (CVAT) reachable by the whole team.

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8. Interface and contract requirements (plan §4.2–§4.4)

ID	Requirement	Pri	Verify
IF-SPEC-1	Both spec forms shall be defined as Pydantic models and validated on load. A spec that does not validate shall not run.	M	T
IF-SPEC-2	The DAG form shall carry: `procedure`, `kind: "dag"`, `state_model`, baseplate dimensions, and a `steps` array as specified in FR-CMP-2.	M	T
IF-SPEC-3	The constraint form shall carry: `procedure`, `kind: "constraints"`, `state_model`, geometry (e.g. `tower: {layers, slots_per_layer}`), an `actions` array with `preconditions` and `effects`, and a `terminal` array.	S5	T
IF-SPEC-4	Every precondition shall carry a machine `expr`, a human `text` and a `violation` code; alerts shall surface the `text` and log the `violation`.	S5	T
IF-SPEC-5	Specs shall be versioned; a spec file shall record the compiler version and model that produced it.	S	I
IF-EVT-1	Events shall conform to a single schema: `t` (seconds since session start), `source` (`state` | `temporal`), `kind` (`ADD` | `REMOVE` | `STEP_START` | `STEP_END`), optional `part`, optional `pose`, and `confidence`.	M	T
IF-EVT-2	The checker shall accept events from the state tracker and the temporal model through the same entry point, distinguished only by the `source` field.	M	T
IF-STA-1	Every state model shall implement `current()` and `diff()` and nothing procedure-specific shall leak past that boundary.	M	I, T
IF-WS-1	The WebSocket shall carry overlays, checklist/legality status and alerts **outbound only** on the demo path; video frames shall not travel inbound during a demo.	M	I, A
IF-WS-2	The WebSocket message schema shall be documented and versioned so the UI and backend can be developed independently.	S	I
IF-LOG-1	The session log shall be a machine-readable file sufficient to regenerate the alert timeline and the LLM report without the video.	M	T

9. Evaluation acceptance criteria (plan §7)

These are the pass/fail gates for week 11. They are fixed **now**, before results exist.

ID	Question	Metric	Target	Pri
EV-DET-1	Are parts detected where they are?	Median centroid error in studs	≤ 0.4 stud	M
EV-DET-2	"	p95 centroid error in studs	≤ 1.0 stud	M
EV-DET-3	"	mAP@0.5	Secondary/reported only — **not** a gate	M
EV-STA-1	Is the world state right? (Model A)	Placement accuracy, type + pose exact, per step	≥ 90%	M
EV-STA-2	Is the world state right? (Model B)	Same	≥ 85%	S3
EV-STA-3	Is the Jenga lattice right?	Slot occupancy accuracy vs hand-labelled state	≥ 95%	S4
EV-SEG-1	Are steps recognized?	Frame accuracy, Edit score, F1@{10, 25, 50}	Within reach of published numbers on the chosen benchmark	M
EV-ERR-1	Are errors caught? (Model A)	Precision at fixed recall, **per error type**	Precision ≥ 0.8 at recall ≥ 0.8	M
EV-ERR-2	Are errors caught? (Model B)	Same	Precision ≥ 0.7 at recall ≥ 0.7	S3
EV-ERR-3	Is it quiet when nothing is wrong?	False alerts per fully correct build (Model A)	≤ 1.0	M
EV-ERR-4	How early?	Median delay from error to alert	< 2 s	M
EV-LAT-1	Is it usable live?	Per-stage latency vs the §4.1 budget	≥ 15 FPS, per-frame ≤ 40 ms	M
EV-CMP-1	Is the compiler right?	Step accuracy over the eval corpus (n = 15–20)	≥ 0.9	M
EV-CMP-2	"	Dependency accuracy	≥ 0.8	M
EV-CMP-3	"	Pose and constraint accuracy	Reported with the §5.4 error taxonomy	M
EV-JEN-1	Do Jenga rules hold?	Violation precision/recall **per rule**	Precision ≥ 0.8 at recall ≥ 0.8	S5
EV-JEN-2	"	Collapse detection	Reported	S5

9.1 Required baselines

ID	Baseline	Pri
EV-BL-1	**Temporal model only** — no world-state tracker; errors decided from recognized actions alone.	M
EV-BL-2	**End-to-end VLM** — sampled frames plus the manual, asked to list errors. Reported **as measured**; no outcome is predicted in advance, and a split result (VLM better on missed step, worse on position and timing) is a legitimate and interesting finding.	M
EV-BL-3	**Published results** from the benchmark dataset's own paper.	M

9.2 Required ablations

ID	Ablation	Pri
EV-AB-1	Without the world-state tracker (actions only).	M

ID	Ablation	Pri
EV-AB-2	Without the temporal model (state only).	M
EV-AB-3	**Occlusion-window slice** — errors that begin and end inside an occlusion window, state-only vs fused. This is the specific, pre-registered hypothesis of plan §2.5 and the only place the temporal model is expected to win.	S
EV-AB-4	LLM-generated spec vs hand-written spec.	M
EV-AB-5	Zero-shot open-vocabulary detector vs fine-tuned detector.	M
EV-AB-6	With vs without soft warnings, measured on false-alarm rate.	S2
EV-AB-7	Top-down only vs top-down + side camera, on Model B height accuracy.	S1
EV-AB-8	Tier 1 vs tier 2 vs tier 3 generality (plan §1.2).	S

9.3 Generality ladder — reporting requirement

ID	Requirement	Pri
EV-GEN-1	**Tier 1** — unseen LEGO assembly and unseen manual, same bricks and detector. Supports: compiler and checker generalise across assemblies.	S3
EV-GEN-2	**Tier 2** — Jenga tower *build*: unseen domain, new part vocabulary, same graph form and checker. Supports: the perception stack is retargetable.	S4
EV-GEN-3	**Tier 3** — Jenga *game*: unseen procedure topology (constraints, not order), same checker interface. Supports: the spec format itself is general. **This is the claim worth making.**	S5
EV-GEN-4	Tier 1 shall not be presented alone as the generality result; the report shall state the ladder honestly and shall not overclaim.	M

10. Milestone gates (plan §8)

ID	Week	Gate — the requirement is met when…	Pri
MG-1	1	Related-work summary written; dataset access requested and licences verified (DR-PUB-1); hardware in hand (§6); LLM API question settled; environment pinned (SW-E1).	M
MG-2	2	Valid specs for Model A and Model B; spec schema (both forms) committed; compiler v1 + validator running; **eval corpus committed** (DR-CRP-1).	M
MG-3	3	Bricks detected and mapped to studs live; 300+ frames labelled; detector v1 trained.	M
MG-4	4	Correct live build state for Model A; teach-by-demo recorder working; **first Model A runs banked** (DR-LEG-6).	M
MG-5	5	Live alerts on Model A for missed / out-of-order / wrong brick; benchmark features precomputed; first segmentation numbers. *(Temporal box 1 of 2.)*	M
MG-6	6	**End-to-end v0** demo runs; latency budget table filled and measured (NFR-PERF-*); mid-term review passed.	M
MG-7	7	Model A recording topped up; Model B test runs recorded; compiler accuracy table complete over the corpus (EV-CMP-*).	M
MG-8	8	Labels exported; **splits fixed and frozen** (NFR-REP-2); detector v2; **side camera integrated** and Model B heights correct (FR-CAL-5, FR-STA-5).	S1
MG-9	9	Occlusion-window fusion measured (EV-AB-3). **Jenga go/no-go gate decided against must-have status** — if green, Jenga rig, detector and lattice state stand up; if red, Jenga is formally deferred and recorded as such. *(Temporal box 2 of 2.)*	S4
MG-10	10	Remaining LEGO rules in (position, rotation, extra/removed); soft warnings; LLM report; Jenga constraint compiler and checker firing.	S2
MG-11	11	Full evaluation complete: benchmark, Model A, Model B, Jenga, all baselines, all ablations, latency.	M
MG-12	12	UI polished (checklist, alerts, ghost overlay, spec editor, Jenga legality panel); someone outside the team can operate it; demo video recorded.	S6
MG-13	13	**Reserved slack — not allocated to features.** Absorbs overruns from any earlier week. Stretch goals only if genuinely clear.	—
MG-14	14	Report, poster, demo video and final presentation submitted.	M

Gate rule: MG-9 is a hard decision point. Jenga work shall not begin before the must-have set (all **M** requirements) is demonstrably green, and Jenga shall never be promoted to must-have.

11. Deliverable requirements (plan §11)

ID	Deliverable	Requirement	Pri
DL-1	Code	Public repository with setup instructions, trained weights and evaluation scripts.	M
DL-2	Dataset	THIRDEYE-LEGO and THIRDEYE-JENGA videos, specs and labels, released subject to DR-REL-2. Compiler eval corpus released unconditionally.	S
DL-3	Report	Written as a short paper: problem, related work, method, results, ablations, the tier 1–3 generality table, limitations.	M
DL-4	Demo video	2–3 min: a correct Model A build with the checklist ticking; each error type caught live; Model B from its manual alone; a Jenga game with violations called in real time.	M
DL-5	Live demo	Baseplate, cameras, bricks and a Jenga tower. A panel member builds Model B and tries to break it by skipping or swapping steps, then plays Jenga and tries to cheat.	M
DL-6	Publication	Submission to a student research track or a workshop on egocentric/procedural video.	C

12. Constraints, assumptions and exclusions

12.1 Constraints

ID	Constraint
CON-1	14 calendar weeks, with week 13 reserved as slack and not available for features.
CON-2	Team of 3 or 4 (open item OQ-1). With 3, the Perception and App roles merge.
CON-3	One consumer GPU, or free-tier cloud GPUs. All model choices must respect this.
CON-4	Negligible budget: bricks, a Jenga set, cameras, a stand, printed markers, and ≈20 LLM API calls.
CON-5	Fixed cameras only. No head-mounted or moving camera for our own recordings.
CON-6	All visual judgment comes from CV models; every alert comes from an explicit rule. The LLM is text-in, text-out.

12.2 Assumptions

ID	Assumption	If false
ASM-1	Lighting is even and the mat is matte during recording and demo.	Detector precision drops; more labelled frames and lighting control needed.
ASM-2	One person works in the workspace at a time.	Out of scope (§12.3); state attribution breaks.
ASM-3	Public dataset access is granted within a few weeks.	Fall back to an alternative benchmark (DR-PUB-3); own data alone still carries the demo.
ASM-4	Two orthogonal side views give full observability of all 54 Jenga slots, because each layer is end-on to exactly one camera.	Jenga state becomes partially unobservable; tier 3 result is at risk.
ASM-5	Standard 75 × 25 × 15 mm Jenga blocks provide adequate scale reference.	Add an explicit scale marker to the rig.
ASM-6	~150 labelled frames suffice for the single-class Jenga detector.	Label more; Jenga's cheapness argument weakens but does not break.

12.3 Explicitly out of scope

ID	Excluded	Pri
OOS-1	Tiny pieces (1×1 plates, Technic pins), hinged or angled parts.	W
OOS-2	Multiple people building simultaneously.	W
OOS-3	Head-mounted (moving) cameras for our own data.	W
OOS-4	Jenga stability or lean estimation — legality is checked, physics is not.	W
OOS-5	VLM reading picture-only official LEGO instruction pages — **demoted**: a separate research problem with its own failure modes, and Jenga now carries the generality claim for less work. Revisit only if weeks 12–13 are clear.	C

12.4 Stretch items

ID	Item	Pri
STR-1	Synthetic training data from LDraw models rendered in Blender (random poses, lighting, backgrounds).	C
STR-2	Laptop-CPU or Jetson edge deployment.	C
STR-3	Voice alerts.	C

13. Open items blocking requirement finalisation (plan §12)

Each of these leaves a requirement provisional. All should be closed in week 1 unless noted.

ID	Open question	Blocks	Due
OQ-1	Team size: 3 or 4?	CON-2, §2 role assignment	Week 1
OQ-2	Which bricks do we already have, and what must we buy?	HW-5, DR-LEG-3	Week 1
OQ-3	Final designs for Model A (flat, 2 layers) and Model B (3–4 layers).	DR-LEG-1, DR-LEG-2, MG-2	Week 2
OQ-4	What GPU is available: local card, college lab, or Colab/Kaggle only?	NFR-PORT-1, NFR-PORT-2, SW-E2	Week 1
OQ-5	Course review dates, to line up the timeline and the week-9 Jenga gate.	§10 (all MG)	Week 1
OQ-6	How many cameras can we actually mount and synchronise? (2 minimum for Jenga, 3 ideal.)	HW-C1, HW-C2, FR-CAL-6	Week 1
OQ-7	~~Is an LLM API allowed or budgeted?~~ **Closed** — ≈20 calls, cost negligible, Ollama fallback exists; confirm the account in week 1.	NFR-COST-1, SW-8	Closed

14. Traceability

14.1 Requirement class → plan section

Requirement class	Plan section
FR-CMP	§2.1, §4.2, §4.3
FR-CAL	§2.2
FR-PER	§2.3, §4.6
FR-STA	§2.4
FR-TMP	§2.5, §4.7
FR-CHK	§2.6, §4.4
FR-UI, FR-EXP	§2.7, §4.5
NFR-PERF	§3.1
DR-*	§5
HW-*	§3.1, §2.2
SW-*	§3, §4.1
IF-*	§4.2, §4.3, §4.4
EV-*	§7
MG-*	§8
DL-*	§11
CON, ASM, OOS, STR	§6, §10
OQ-*	§12

14.2 Risk → mitigating requirement (plan §10)

Risk	Likelihood	Mitigated by
Height/layer wrong from top-down view	**High**	DR-LEG-1 (flat Model A), FR-CAL-5, FR-STA-5, HW-8, MG-8, EV-AB-7
Similar bricks confused (red/orange, 2×3/2×4)	High	DR-LEG-3 (distinct vocabulary), DR-LEG-8 (more labels), STR-1
Hands block the view during placement	High	FR-PER-6 (freeze), FR-STA-3 (stability filter), FR-TMP-5, EV-AB-3
Labelling takes longer than planned	High	DR-LEG-9 (30 h budget), DR-LEG-8 (boundaries only), FR-CMP-8 (states from demo)
Jenga scope creep sinks the must-have	Medium	MG-9 (go/no-go gate), priority scheme (§1.5): Jenga is never **M**
Jenga far-side blocks unobservable	Medium	FR-CAL-6, HW-9, ASM-4
Step recognition too inaccurate	Medium	FR-TMP scope note (benchmark-only), FR-STA-4 (demo relies on state diffs)
Too many false alarms	Medium	FR-CHK-9 (soft warnings), FR-STA-3, NFR-REL-2 and EV-ERR-3 (explicit budget)
Dataset access slow or denied	Medium	DR-PUB-1, DR-PUB-3 (alternatives kept viable)

Risk	Likelihood	Mitigated by
Not enough GPU	Medium	FR-TMP-1 (precomputed features), SW-3 (small detector variant), NFR-PORT-1
LLM specs wrong	Low	FR-CMP-6 (validator), FR-CMP-7 (one repair round), FR-CMP-8, FR-CMP-9, EV-CMP-*
Jenga tower collapses mid-run	Certain	FR-STA-8 (terminal event, auto-saved clip) — a demo highlight, not a failure

14.3 Must-have set (the minimum viable project)

The project passes its minimum bar when every **M** requirement is verified. In plan terms (§6): the compiler produces a validated dependency spec; calibration, detection, tracking and the world-state tracker work; the checker catches **missed step**, **out of order** and **wrong brick**; the live demo runs at 15+ FPS within the latency budget; the step recognizer is trained and benchmarked against at least one baseline; and compiler accuracy is measured on the eval corpus.

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15. Change control

ID	Requirement
CC-1	Requirement IDs are stable. A requirement that is dropped is marked **superseded** with a reason; it is never deleted or renumbered.
CC-2	Any change to an **EV-** target after week 11 results exist must be recorded as a change with a date and a rationale, and the original target reported alongside.
CC-3	Promotion of any **S** or **C** item to **M** requires the must-have set to be green first (MG-9 rule).
CC-4	This document is regenerated from the plan whenever the plan changes materially; the plan is the source of truth.

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